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Review

Divergence of pheromone communication systems in noctuid moths: From biosynthesis to recognition

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Abstract Sex pheromones are central to mate recognition in moths and often contribute to premating isolation and evolutionary divergence. In moths, female-produced pheromone blends are typically highly species-specific, and even subtle variation in blend composition or component ratios can alter male attraction and reinforce reproductive barriers between closely related species. Recent advances in genomics, functional genetics, and receptor characterization have substantially enhanced our understanding of how pheromone communication systems diverge in sympatric moths through the coevolution of female signals and male perception. Noctuid moths, with their well-documented pheromone production and communication systems, provide an excellent model for studying the evolution of species-specific mate recognition. Here, we review recent advances in the study of pheromone communication in noctuid moths, focusing on three interconnected aspects: pheromone biosynthesis, receptor-mediated recognition, and the evolutionary processes underlying communication divergence. We also assess the extent to which current data support a link between pheromone divergence, male preference, and premating reproductive isolation. Although direct empirical evidence connecting molecular changes to long-term lineage splitting remains limited, noctuid moths provide a powerful comparative system for investigating how chemical communication evolves and contributes to reproductive divergence in insects. More broadly, they offer a tractable model for understanding how signal-receiver coevolution and gene-family evolution interact during premating isolation and lineage divergence in chemically communicating animals.

Key words: divergence, noctuid moths, pheromone communication, pheromone receptor, premating isolation, sex pheromone.

1 Introduction

Mate recognition is crucial for facilitating communication between the sexes in insects and relies heavily on sex pheromones (Cardé & Baker, 1984). In lepidopteran moths, females typically release pheromones detected by males, forming a species-specific communication system that promotes reproductive isolation (Cardé & Haynes, 2004). Male moths exhibit high sensitivity to their conspecific sex pheromone blend, which usually comprises a combination of structurally related molecules (Ando et al., 2004). These pheromone signals are likely under strong stabilizing selection, as closely related species often share similar blends that differ primarily in qualitative traits (e.g., presence or absence of components, double bond or enantiomer configurations, chain length, and functional moieties) or quantitative traits (e.g., component ratios) (Allison & Cardé, 2016). For mating success, the signals emitted by females and detected by males must be precisely matched (Linn & Roelofs, 1983). Even slight deviations in release rates or component ratios can significantly affect male behavior (Löfstedt, 1993; Roelofs et al., 2002). This raises an important question: How do shifts in female pheromone production and corresponding male preferences coevolve to establish new, species-specific attraction patterns? Addressing this challenge experimentally is difficult, as the establishment of divergent communication systems typically occurs over long timescales, complicating the identification of underlying mechanisms. Nonetheless, a thorough understanding of the genetic and evolutionary processes driving divergence in pheromone composition and recognition across closely related taxa could provide crucial insights.

Noctuidae, one of the largest families within Lepidoptera, comprises more than 12 000 described species (van Nieukerken, 2011). These moths are distributed across all major continents except Antarctica and thrive in diverse habitats, ranging from hydric to ultra-xeric conditions, including many oceanic islands (Keegan et al., 2021). Noctuids represent a significant portion of global biodiversity and include some of the most destructive agricultural pests, such as species within the genera *Heliothis*, *Helicoverpa* (Cho et al., 2008), *Spodoptera* (Parra et al., 2022), and *Mythimna* (Jiang et al., 2011). To develop control strategies for these widely distributed and agriculturally important pests, their pheromone recognition systems have been a major research focus over the past few decades. Recent advances in functional genomics have opened new avenues for identifying and characterizing the genes involved in pheromone biosynthesis and detection in this group. Noctuid moths are among the most extensively studied insects regarding their pheromone communication systems. Their well-characterized biology and identified pheromone receptor (PR) genes make them ideal models for investigating how the evolution of pheromone communication contributes to species-specific mate recognition, premating isolation, and broader evolutionary divergence (Wu, 2001). Insights from noctuid moths are also relevant beyond this family, as they address general evolutionary questions concerning the origins of communication system diversity, the emergence of premating barriers, and how lineage-specific molecular changes correspond to broader patterns of species divergence.

2 Biosynthesis and evolution of sex pheromones in noctuid moths

2.1 Sex pheromones of noctuid moths

Since the identification of the first insect pheromone, *E*10, *Z*12-hexadecadiene-1-ol (E10,Z12-16:OH), in the silkmoth *Bombyx mori* (Butenandt, 1959), the sex pheromones of approximately 2000 lepidopteran moth species have been investigated (Jurenka, 2021). Based on chemical structure, moth sex pheromones are categorized into two primary groups (types I and II) and two additional miscellaneous groups (types 0 and III). Type I pheromones, typically monounsaturated or diunsaturated acetates, alcohols, or aldehydes with 10–18 carbon atoms, account for about 75% of known moth pheromones (Ando et al., 2004). These pheromones are synthesized *de novo* in the pheromone glands of female moths, specialized epidermal tissues located near the terminal abdominal segments (Löfstedt et al., 2016). Type II pheromones, consisting mainly of polyunsaturated hydrocarbons or their epoxide derivatives with unbranched carbon chains of C17–C25 (Roelofs et al., 1982), constitute approximately 15% of known moth pheromones and are derived from diet-based linoleic or linolenic acids (Ando et al., 2004). Type III pheromones include both saturated and unsaturated hydrocarbons, as well as functionalized hydrocarbons with one or more methyl branches (Bierl et al., 1970). Type 0 pheromones, identified in two primitive moth lineages, are short-chain methylcarbinols and methylketones and represent an ancestral form of moth pheromones (Larsson et al., 2002). Although some moth pheromones do not fit neatly into these four categories (Hall et al., 1987), the major pheromone types encompass most of the chemical diversity observed in moth sex pheromones (Löfstedt et al., 2016).

Until recently, sex pheromone components have been identified in approximately 380 noctuid moth species, with nearly all exhibiting type I sex pheromones, except for two species (El-Sayed, 2024). *Tmetolophota atristriga* possesses a hybrid pheromone system combining both Type I and Type II components (El-Sayed & Manning, 2022), while *Aedia leucomelas* exhibits type II sex pheromones (Choi et al., 2004). A phylogenetic study by Keegan et al., based on eight protein-coding genes, revealed that *A. leucomelas* is more closely related to heliothine species (Keegan et al., 2021), suggesting that Type II sex pheromones in noctuid moths evolved independently and that similar components may exist in other noctuid species. Recent evidence supporting this hypothesis comes from a study by Wang et al., which identified a Type II pheromone compound in the abdomen of female *Helicoverpa armigera* (Wang et al., 2024b). Although the behavioral impact of this compound remains unclear, electrophysiological results indicate that both male and female moths can detect it. Further research is necessary to determine whether similar compounds are present in the female sex glands or abdomens of other noctuid species and whether they influence mating behavior.

Despite extensive research on the sex pheromone components of various moth species, substantial discrepancies are often observed in both the quantity and composition of pheromones reported for the same species, particularly in well-studied taxa. For example, although the primary sex pheromone component of *Spodoptera frugiperda* (Z9-14:Ac) is consistently reported, significant differences exist in the minor components across studies (El-Sayed, 2024). Several factors likely contribute to these discrepancies, including variations in pheromone collection methods (e.g., moth age, timing of extraction, and environmental conditions) and differences in the sensitivity and precision of GC-MS instruments. Researcher experience and data analysis approaches may further influence results. Geographic variation in sex pheromone composition is also common; for instance, populations of *Agrotis segetum* from Denmark, France, Hungary, and Switzerland exhibit distinct pheromone profiles (Löfstedt et al., 1986), as do populations of *H. armigera* (Gao et al., 2020), *Heliothis virescens*, *H. subflexa* (Groot et al., 2009b), and *Mythimna separata* (Zhu et al., 1987; Fónagy et al., 2011). This variation may reflect adaptations that reduce communication interference with sympatric, closely related species or broader ecological influences, such as host-associated environments and other conditions affecting pheromone production or response (Groot et al., 2006; Mochizuki et al., 2008; De Pasqual et al., 2021). These findings highlight the flexibility of sex pheromone composition and suggest that intraspecific variation can play a key role in the evolution of moth pheromone communication and contribute to lineage divergence.

Once pheromone components are identified from female pheromone glands or other tissues, electrophysiological and behavioral studies are essential to confirm whether these compounds function as true sex pheromones. This is particularly important because many compounds classified as pheromones may not meet stringent criteria for true sex pheromones (Smart et al., 2014). For instance, Zhang et al. identified 10 candidate pheromone compounds in the female pheromone glands of *H. armigera*, but several failed to elicit a behavioral response in males (Zhang et al., 2012a). This phenomenon is commonly observed in other moth species as well. However, investigating the behavioral effects of individual components on conspecific or heterospecific adults is challenging due to numerous variables, including pheromone blend composition, compound concentration, volatility, adult moth availability, and geographic population differences. Nonetheless, when feasible, such studies can deepen our understanding of each pheromone component's function and its role in communication divergence while also enhancing the application of these compounds in pest control.

2.2 The sex pheromone biosynthetic pathways in noctuid moths

Noctuid moths primarily utilize type I pheromones as their sexual communication signals (El-Sayed, 2024), which are synthesized in specialized pheromone glands located between the final and penultimate abdominal segments (Rafaeli, 2009). Pheromone release is regulated by photoperiodic cues that trigger the secretion of pheromone biosynthesis-activating neuropeptide (PBAN) from the brain-subesophageal ganglion complex (Raina et al., 1989). PBAN is then released into the hemolymph, where it activates a G-protein-coupled receptor (PBANr) in the pheromone gland (Kim et al., 2008), leading to the opening of calcium channels. This results in Ca^{2+} influx and cyclic AMP (cAMP) production, which act as secondary messengers to activate specific enzymes involved in pheromone biosynthesis (Jurenka et al., 1991).

A common approach to studying pheromone biosynthesis involves identifying candidate genes through transcriptome or genome sequencing of the target species. Several key enzyme families involved in pheromone production have been identified and functionally characterized in noctuid moths, including acetyl-CoA carboxylase (ACC), fatty acid synthase (FAS) enzymes, desaturases, chain-shortening enzymes, fatty acyl reductases (FAR), and enzymes that modify functional groups on the carbonyl carbon (Jurenka, 2021). ACC converts acetyl-CoA to malonyl-CoA, which FAS enzymes then use to elongate carbon chains to 16:CoA and 18:CoA. These long-chain fatty acids can be shortened by oxidase enzymes, removing two carbon units per cycle. Desaturases are crucial for introducing double bonds into saturated or monounsaturated fatty acid precursors, significantly contributing to the structural diversity of moth sex pheromone components. These enzymes exhibit diverse substrate specificities, regiospecificities, and stereospecificities, enabling the production of unsaturated fatty acid precursors with varying chain lengths, double bond positions, and both *Z* (*cis*) and *E* (*trans*) isomers (Knipple et al., 2002). Multiple desaturases have been identified in moths, including Δ5 (Hagström et al., 2014), Δ6 (Wang et al., 2010), Δ9 (Zhang et al., 2019), Δ10 (Foster & Roelofs, 1988), Δ11 (Serra et al., 2006), Δ12 (Xia et al., 2019), Δ13 (Villorbina et al., 2003), and Δ14 desaturases (Roelofs et al., 2002). However, only Δ9, Δ11, and Δ12 desaturases have been identified in noctuid moths. The chain-length CoA thioester intermediates generated by desaturases and chain-shortening enzymes are converted into alcohols by fatty acyl-CoA reductase (pgFAR) in the pheromone gland. These alcohols may serve as pheromone components themselves or be converted into aldehydes and acetate esters by alcohol oxidase and acetyl transferase, respectively. Alternatively, aldehydes and acetate esters can be reconverted into alcohols through aldehyde reductase and acetate esterase.

Here, we summarize the potential biosynthetic pathways for sex pheromone production in female moths from the genera *Helicoverpa* and *Spodoptera* (Fig. 1). In *Helicoverpa*, sex pheromone components in most species include Z11-16:Ald and Z9-16:Ald (Hillier & Baker, 2016). Detailed biosynthetic pathways for these components have been elucidated in *H. armigera*, *H. zea*, and *H. assulta* using stable isotope-labeled precursors and GC/MS analysis (Choi et al., 2002; Wang et al., 2005). The biosynthesis of Z11-16:Ald in *Helicoverpa* involves Δ11 desaturation of 16:CoA to form Z11-16:CoA, which is then reduced to the corresponding alcohol before conversion to the aldehyde. In contrast, Z9-16:Ald can be synthesized either by Δ11 desaturation of 18:CoA to form Z11-18:CoA, followed by chain-shortening to Z9-16:CoA, or via Δ9 desaturation using 16:CoA as the substrate (Choi et al., 2002; Wang et al., 2005).

The pheromone structures in *Helicoverpa* differ from those in certain *Spodoptera* species, which rely on a more complex blend of acetate esters containing both monounsaturated and diunsaturated components. In *Spodoptera*, the conjugated diene Z9,E11-14:CoA is synthesized through Z11 desaturation of 16:CoA, followed by chain-shortening to Z9-14:CoA and subsequent E11 desaturation to form Z9,E11-14:CoA (Rodríguez et al., 2004; Xia et al., 2019). Alternatively, E12 desaturase can produce the unconjugated Z9,E12-14:CoA (Dunkelblum & Kehat, 1987). Additionally, the conjugated diene E10,E12-14:CoA is produced by Z11 desaturation of 14:CoA, followed by Δ9 desaturation, which rearranges the double bonds to yield E10,E12-14:CoA (Martinez et al., 1990; Muñoz et al., 2008). These biosynthetic pathways in *Helicoverpa* and *Spodoptera* provide valuable insights into pheromone biosynthesis in noctuid moths and offer a mechanistic foundation for interpreting signal divergence among closely related species.

2.3 Divergence in the composition and function of sex pheromones in closely related noctuid moths

Female moths produce species-specific sex pheromone blends in precise component ratios to attract conspecific males. For sympatric, closely related species, avoiding mating errors during mate finding and courtship is crucial (Smadja & Butlin, 2009). Male moths are highly sensitive and selective, responding to precise multicomponent mixtures of female-emitted pheromones. Any deviation in the ratios of key components can significantly affect female attraction to conspecific males and avoidance of heterospecifics. Extensive studies have examined sex pheromone components and their functional roles across closely related species, particularly within the genera *Heliothis*, *Helicoverpa*, and *Spodoptera*, highlighting how variation in blend composition and function contributes to communication divergence and premating isolation.

The closely related moth species *H. virescens* and *H. subflexa* occur sympatrically across North and South America and exhibit overlapping nocturnal activity periods (Heath et al., 1991). Both species share the same major pheromone component, Z11-16:Ald, which is essential for attracting conspecific males. However, they differ markedly in the composition and ratios of minor pheromone

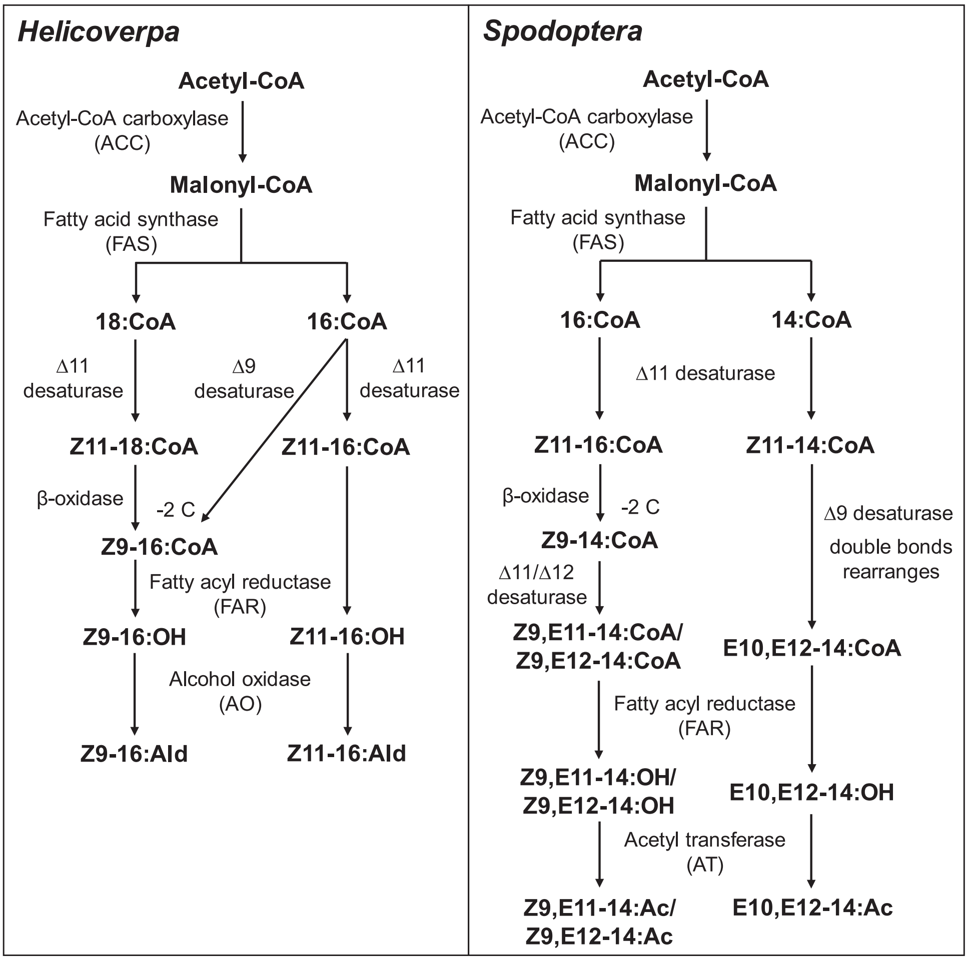

Fig. 1. Schematic drawing of the possible sex pheromone biosynthetic pathways in female moths of *Helicoverpa* and *Spodoptera*.

components (Fig. 2). In *H. virescens*, the primary secondary component is Z9-14:Ald, whereas *H. subflexa* relies on a blend including Z9-16:Ald, Z11-16:OH, and Z11-16:Ac, all required for effective conspecific male attraction (Teal et al., 1981; Vetter & Baker, 1983; Teal et al., 1986; Vickers, 2002; Groot et al., 2005). When these minor components from *H. subflexa* are added to the *H. virescens* pheromone blend, they elicit strong repellent responses in *H. virescens* males (Vickers & Baker, 1997; Groot et al., 2009a). This reciprocal behavioral response to identical chemical cues indicates that the two sympatric *Heliothis* species are strongly differentiated at the premating stage, primarily through differences in minor pheromone components.

Four *Helicoverpa* moths—*H. assulta*, *H. armigera*, *H. zea*, and *H. gelotopoeon*—have overlapping geographic distributions, with the former two sharing regions across Africa, Asia, and Oceania and the latter three overlapping in South America (Hillier & Baker, 2016). These species differ in major pheromone components: *H. assulta* uses Z9-16:Ald; both *H. armigera* and *H. zea* use Z11-16:Ald (Klun et al., 1980; Pope et al., 1984; Kehat & Dunkelblum, 1990; Zhang et al., 2012a), and *H. gelotopoeon* uses 16:Ald and Z9-16:Ald (Cork et al., 1992; Cork & Lobos, 2003) (Fig. 2). In addition to minor components, behavioral effects exhibit significant divergence among the species (Fig. 2). Z11-16:OH, identified in *H. armigera* pheromone glands, antagonizes *H. armigera* and *H. zea* males but does not affect *H. assulta* or *H. gelotopoeon* (Quero & Baker, 1999; Chang et al., 2017). Trace amounts of Z9-14:Ald (~0.3%) in the *H. armigera* pheromone blend enhance attraction to *H. armigera* and *H. zea* males but exhibit behavioral antagonism at high doses and also antagonize *H. assulta* males when added to their optimal blend (Boo et al., 1995; Zhang et al., 2012a; Wu et al., 2015). In contrast, *H. gelotopoeon* uses Z11-16:Ald as a pheromone antagonist (Cork & Lobos, 2003).

Four *Spodoptera* species—*S. exigua*, *S. littoralis*, *S. litura*, and *S. frugiperda*—have received considerable attention regarding their sex pheromones and associated behavioral responses (Fig. 2). Several occur sympatrically in different regions: *S. littoralis* and *S. frugiperda* in Africa and Asia; *S. litura*, *S. exigua*, and *S. frugiperda* in Asia; and *S. litura* and *S. frugiperda* in Oceania (EPPO Global Database, https://gd.eppo.int/search; Zheng et al., 2011). The pheromone blends essential for conspecific male attraction differ substantially among these species in both component composition and behavioral function (Fig. 2). Although both *S. littoralis* and

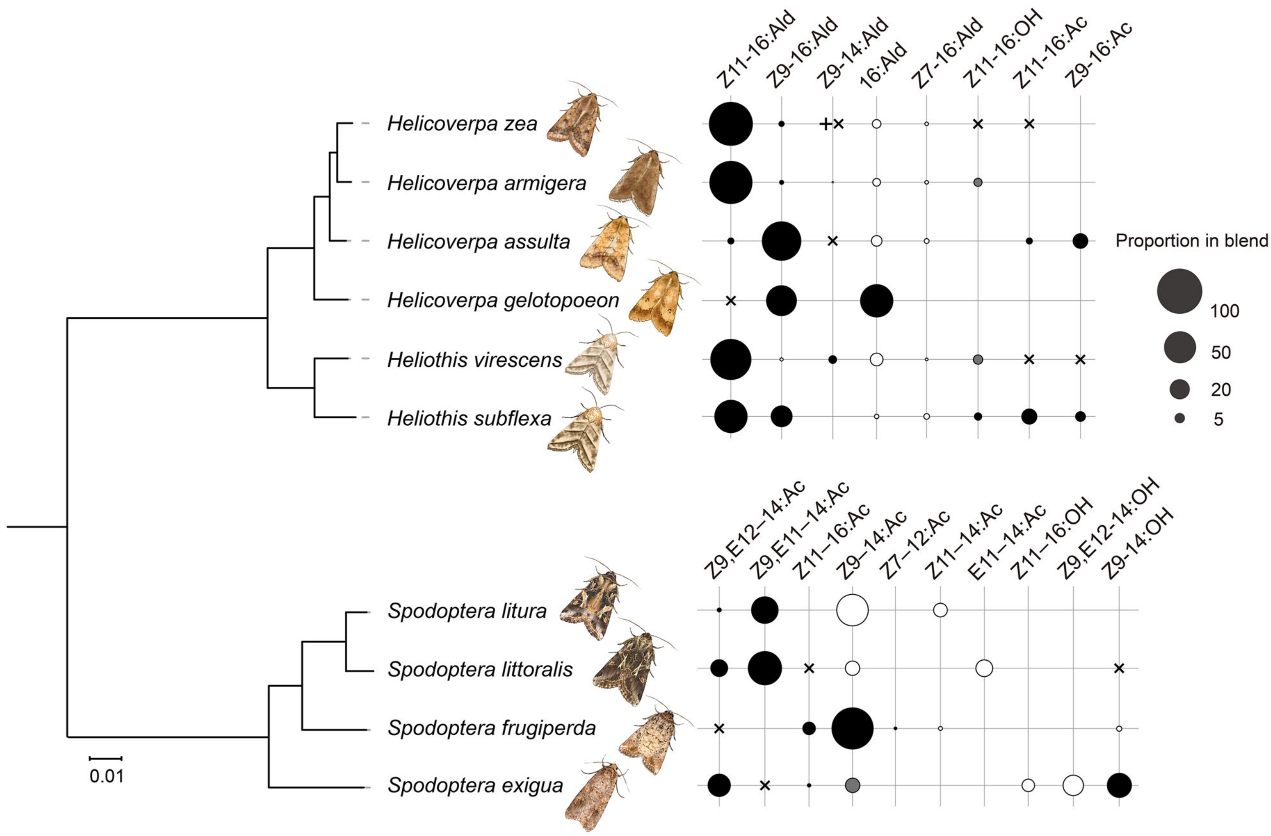

Fig. 2. Compositions and behavioral effects of female sex pheromones in representative heliothine and *Spodoptera* moth species. The phylogenetic tree was constructed as described in Table S1. The size of each circle is proportional to the relative ratio of each individual pheromone component (Table S1). Black-filled circles indicate compounds identified in the female pheromone blend that attract conspecific males, dark-gray-filled circles indicate compounds in the female pheromone blend that repel males, and open circles with black outlines indicate compounds in the female pheromone blend with no detectable effect on intraspecific male behavior. Black plus signs and black crosses indicate compounds not identified in the female pheromone blend but showing attractive or repellent effects on males, respectively.

S. litura use Z9,E11-14:Ac as their major component and Z9,E12-14:Ac as their minor component (Tamaki & Yushima, 1974; Dunkelblum et al., 1982; Dong & Du, 2002; Sun et al., 2002), their geographic distributions do not overlap. Furthermore, Z9,E11-14:Ac is absent in *S. exigua* and *S. frugiperda* and inhibits *S. exigua* male attraction to its pheromone blend (Yan et al., 2019). Z9,E12-14:Ac is a common pheromone component except in *S. frugiperda*, where it acts as an attractant inhibitor (Mitchell et al., 1974). Similarly, Z9-14:OH and Z11-16:Ac are essential in the *S. exigua* pheromone blend but strongly inhibit *S. litura* male catches to its conspecific sex pheromone (Yan et al., 2019). The major component in *S. frugiperda*, Z9-14:Ac (Wakamura et al., 2020; Wang et al., 2022), differs considerably from the other three species and significantly reduces *S. exigua* male catches to its sex pheromone (Tumlinson et al., 1990). Other unidentified sex pheromone components may also inhibit cross-attraction between sympatric species.

Two *Mythimna* moth species, *M. separata* and *M. loreyi*, are sympatric in parts of Asia and Australia (Sharma & Davies, 1983; Duan et al., 2022). In *M. loreyi*, female pheromone glands produce Z9-14:Ac, Z7-12:Ac, and Z11-16:Ac in a ratio of 100:8.8:19.7, a blend highly effective in attracting conspecific males (Wang et al., 2023). In contrast, in *M. separata*, Z11-16:Ald, the only compound eliciting electrophysiological responses in pheromone gland extracts, is sufficient to trigger the full sequence of male sexual behaviors (Jiang et al., 2019). These two *Mythimna* species use distinct pheromone blends to attract males, significantly reducing interspecific hybridization potential. Collectively, these examples demonstrate that divergence in pheromone composition and behavioral responses reduces cross-attraction and strengthens premating isolation among closely related noctuid species.

2.4 The genetic basis of sex pheromone variation in noctuid moths

A central question in pheromone divergence concerns how species-specific ratios or novel chemical signals are generated, particularly in closely related moth species with highly similar or identical biosynthetic pathways. Earlier studies employed Quantitative Trait Locus (QTL) analysis by hybridizing closely related species and back-crossing F1 offspring to parental lines. This approach successfully identified key QTLs associated with specific sex pheromone component production in noctuid moths, including

H. virescens and *H. subflexa* (Sheck et al., 2006; Groot et al., 2009a, 2013) and *H. armigera* and *H. assulta* (Wang et al., 2008). Although QTL mapping is effective in some cases, its efficiency is limited. Fortunately, recent advances in genomic technologies have greatly enhanced the ability to identify genes involved in pheromone biosynthesis.

Next-generation sequencing technologies have been used to analyze genomes (Qu et al., 2022; Zhao et al., 2023) or transcriptomes of female moth pheromone glands (Vogel et al., 2010; Zhang et al., 2014, 2017; Ding & Löfstedt, 2015; Li et al., 2017) to identify candidate pheromone biosynthesis genes. These candidate enzymes can be validated through gene cloning and expression in yeast systems (Hagström et al., 2013) to confirm substrate specificity and product output. Additionally, RNA interference (Ohnishi et al., 2006) or gene editing (Shi et al., 2024) can validate the in vivo functions of these enzymes. This integrated approach is broadly applicable for identifying and functionally analyzing enzymes in moth pheromone biosynthesis pathways. Thus, in theory, the function of any enzyme involved in moth pheromone production can be elucidated through this methodology.

Studies on noctuid moths have revealed several mechanisms contributing to pheromone composition variation. First, changes in key enzyme expression levels can alter specific pheromone component ratios. For example, the sibling species *H. armigera* and *H. assulta* share the same two sex pheromone components, Z11-16:Ald and Z9-16:Ald, but in opposite ratios (Wang et al., 2005). Comparative pheromone gland transcriptome analyses suggest that variation in Δ11 and Δ9 desaturase gene expression is likely the primary factor regulating these contrasting ratios (Li et al., 2015; Li et al., 2017).

Second, mutations in genes encoding enzymes within pheromone biosynthetic pathways can alter enzymatic activity, thereby affecting sex pheromone production or ratios. In *H. virescens*, a single mutation in the Δ11 desaturase gene introduces a premature stop codon, resulting in a significant shift in the ratio of two key pheromone components (Groot et al., 2019). Similar findings have been reported in moths from other families (Lassance et al., 2010; Lassance et al., 2013; Buček et al., 2015; Ding et al., 2016).

Third, enzymes involved in pheromone degradation play a crucial role in regulating specific pheromone component levels. In *H. subflexa*, the sex pheromone blend includes three acetate esters, Z11-16:Ac, Z7-16:Ac, and Z9-16:Ac, that are absent in *H. virescens*. These acetate esters enhance attraction to *H. subflexa* males while inhibiting attraction to *H. virescens* males, thereby contributing to premating isolation (Vickers & Baker, 1997; Groot et al., 2007). Recent studies show that two lipases and two esterases, expressed at higher levels in *H. virescens* and involved in acetate hydrolysis, significantly impact acetate ester abundance in the sex pheromone blends of these species (de Fouchier et al., 2023). Overall, species-specific pheromone blends result from the coordinated action of biosynthesis and degradation pathway enzymes. Variations in enzyme expression levels and functional differences can shift pheromone composition and ratios, thereby contributing to communication divergence and reproductive isolation maintenance across species.

Although the studies above highlight the genetic basis of pheromone variation in noctuid moths, female sex pheromones are unlikely to be shaped by genetic factors alone. Evidence from noctuid systems indicates that geographic setting and host-associated environments may also contribute to pheromone composition variation (Groot et al., 2009b; Gao et al., 2020), although the direct molecular pathways linking environmental conditions to biosynthetic regulation remain poorly understood. Thus, pheromone blend divergence in noctuid moths should be viewed not only as the outcome of biosynthetic and degradation gene changes but also as traits whose expression and evolutionary stability may be influenced by ecological context.

3 Mechanisms and evolution of sex pheromone communication in noctuid moths

3.1 Mechanisms of pheromone recognition in moths

Female moths release species-specific sex pheromone components. To locate suitable mates among sympatric species, male moths have evolved remarkably sensitive and selective pheromone communication systems. Pheromone perception in males is a complex process (Sakurai et al., 2014). Volatile pheromone molecules are detected by odorant receptor neurons (ORNs) housed within pheromone-sensitive trichoid sensilla on male antennae. Once pheromone molecules enter the sensilla, they are captured and bound by pheromone-binding proteins (PBPs), which transport them to pheromone receptors (PRs). PRs are activated by pheromone molecules, converting chemical signals into electrical ones. After signal transduction, pheromone molecules are inactivated through enzymatic degradation or other unidentified mechanisms. The signals are then transmitted to the antennal lobe (AL), where ORNs expressing the same odorant receptor converge onto specific glomeruli (Vosshall et al., 2000). Pheromone information is relayed to and processed by higher-order centers, such as the mushroom body (MB) and lateral protocerebrum (LPC), ultimately eliciting male orientation behavior toward the female (Sakurai et al., 2014).

The AL serves as the primary olfactory center and comprises two regions: the macroglomerular complex (MGC) and the ordinary glomeruli (OG) (Galizia et al., 2000; Carlsson et al., 2002). In male heliothine moths, the MGC includes compartments such as the cumulus (Cu), dorsomedial anterior unit (DMA), and dorsomedial posterior unit (DMP) (Wu et al., 2015). These compartments specialize in processing different pheromone signals related to both intraspecific and interspecific communication (Berg et al., 2005; Liu et al., 2023). The AL contains two neuron types: local interneurons (LNs), which connect to specific glomeruli, and projection neurons (PNs), which receive input from glomeruli and relay processed information to higher olfactory centers. In *H. armigera*, the LPC receives pheromone signals via male-specific PNs along three primary tracts: the medial (mALT), mediolateral (mlALT), and lateral ALT (lALT) (Kymre et al., 2021). The MBs, paired structures in the insect brain, are involved in memory formation, sensory integration, and context recognition (Heisenberg, 2003).

In *Spodoptera* moths, the MBs are longitudinally subdivided into distinct neuropil domains, with circuits within the calyces and lobes potentially modulated independently (Sinakevitch et al., 2008).

3.2 Molecular components in pheromone communication system

The pheromone signaling process in moths involves multiple protein families, including PRs (Chang et al., 2017; Bastin-Héline et al., 2019), PBPs (Ye et al., 2017; Zhu et al., 2019), sensory neuron membrane proteins (SNMPs) (Pregitzer et al., 2014; Liu et al., 2020), and pheromone-degrading enzymes (PDEs) (Durand et al., 2011; He et al., 2014). Additionally, in *Drosophila*, ionotropic receptors (IRs) (Koh et al., 2014) and gustatory receptors (GRs) (Watanabe et al., 2011) have been reported to influence sexual behaviors, although it remains unclear whether they directly interact with sex pheromone components. However, similar findings have not yet been reported in moths.

PBPs are a subfamily of the odorant-binding protein (OBP) family in insects, defined by their ability to bind pheromone compounds and their predominant expression in the male antennae of the giant silkmoth *Antheraea polyphemus* (Vogt & Riddiford, 1981). PBPs are small, soluble proteins synthesized by two of the three accessory cells (trichogen or tormogen cells) and secreted into the sensillum lymph at concentrations up to 10 mM (Klein, 1987). In moths, each species typically possesses three PBPs, designated PBP1-3. In *H. armigera*, the three PBPs are localized in the supporting cells of trichoid sensilla, with *HarmPBP1* and *HarmPBP2* colocalized in cells surrounding ORNs expressing *HarmOR13* (Guo et al., 2021). In vitro expression and ligand-binding assays revealed that HarmPBP1 strongly binds Z11-16:Ald and Z9-16:Ald, whereas HarmPBP2 and HarmPBP3 show weaker affinities for tested ligands (Zhang et al., 2012b). Heterologous expression of these PBPs with *HarmOR13* in *Drosophila* T1 sensilla showed that adding HarmPBP1 or HarmPBP2, but not HarmPBP3, significantly enhances HarmOR13′s response to Z11-16:Ald (Guo et al., 2021). In addition to PBPs, a general odorant-binding protein (GOBP) from *S. frugiperda* has been reported to exhibit binding affinities for the four sex pheromones of female *S. frugiperda* (Yang et al., 2023).

SNMP, a homolog of the vertebrate CD36 receptor family, was first identified as an essential molecular component required for pheromone sensitivity in *Drosophila* (Benton et al., 2007; Jin et al., 2009). In Lepidoptera moths, two SNMP genes are generally present: *SNMP1*, which is highly expressed in adult antennae, particularly within pheromone-sensitive ORNs of trichoid sensilla, and *SNMP2*, which is expressed in the accessory cells surrounding the odorant-sensitive ORNs in adult antennae as well as in several other tissues (Forstner et al., 2008; Liu et al., 2014; Zhang et al., 2015a). The crucial role of SNMP1 in pheromone detection has been confirmed in moths (Pregitzer et al., 2014; Liu et al., 2020), while the function of SNMP2 remains poorly characterized. Recently, a novel SNMP gene, *SNMP3*, was identified in Lepidoptera moths as a true homologue of the dipteran *SNMP2*. Moreover, *SNMP3* is highly expressed in the midgut of both larvae and adults (Zhang et al., 2020), although its function still requires further investigation.

PRs are key components of the peripheral pheromone communication system, directly binding to pheromone molecules to initiate signal transduction, making them crucial targets for studying the function and evolution of pheromone communication in moths. PRs are a subset of odorant receptors (ORs) expressed on the dendritic membranes of pheromone receptor neurons (PRNs). They possess a seven-transmembrane structure and are directly activated by pheromone molecules through a unique class of heteromeric ligand-gated ion channels, which include a conserved co-receptor subunit (Orco) in an unexpected stoichiometry of one OR to three Orco subunits (Wang et al., 2024c; Zhao et al., 2024). Since the identification of the first PR genes in *H. virescens* (Krieger et al. 2004) and *B. mori* (Sakurai et al., 2004), genomic analyses over the past two decades have led to the identification of PR genes in a wide range of moth species (Zhang & Löfstedt, 2015). Moth PRs are relatively conserved, typically clustering within the same branch of the phylogenetic tree, forming a distinct clade for traditional moth PRs. This clustering serves as an important criterion for PR gene identification in moths (Zhang & Löfstedt, 2015). Recent studies have revealed that the PR (SlitOR5) in *S. littoralis* responsible for recognizing its major pheromone component, Z9,E11-14:Ac, does not belong to this clade but instead resides in a novel branch (Bastin-Héline et al., 2019). This finding suggests two independent origins of PRs tuned to Type I pheromones in Lepidoptera. Further studies have identified additional PR genes within this novel clade in other moth species, which can also be activated by their sex pheromones or analogs (Shen et al., 2020; Yuvaraj et al., 2022; Li et al., 2023; Zhang et al., 2024). Since sex pheromones are primarily released by female moths and detected by conspecific males, PR genes, in most cases, exhibit specific or biased expression in male antennae (Figs. 3, 4) (Vásquez et al., 2011; Liu et al., 2013a; 2013b; Zhang et al., 2015b; 2023; Yang et al., 2017; Guo et al., 2020; Yang & Wang, 2020; Guo et al., 2022), although exceptions do exist.

3.3 Evolutionary mechanisms underlying pheromone recognition divergence in noctuid moths

Sex pheromones and their recognition systems in moths exhibit a high degree of species specificity. However, the mechanisms driving the evolution and establishment of new communication systems within populations remain poorly understood, largely because the divergence of signal-response systems typically occurs over extended time periods. Noctuid moths, with extensive studies on the identification and functional characterization of PR genes among closely related species, offer an excellent model for exploring the evolutionary dynamics of male pheromone communication systems. Figs. 3 and 4 provide detailed comparisons of gene expression levels and functions of PRs across six heliothine species and four *Spodoptera* species, respectively, illustrating several mechanisms that may underlie pheromone recognition divergence in closely related noctuid moths.

One key mechanism for the evolution of PRs is the birth-and-death process, which drives changes in the number of functional receptors. Large gene families encoding ecologically significant proteins, such as chemoreceptors, evolve

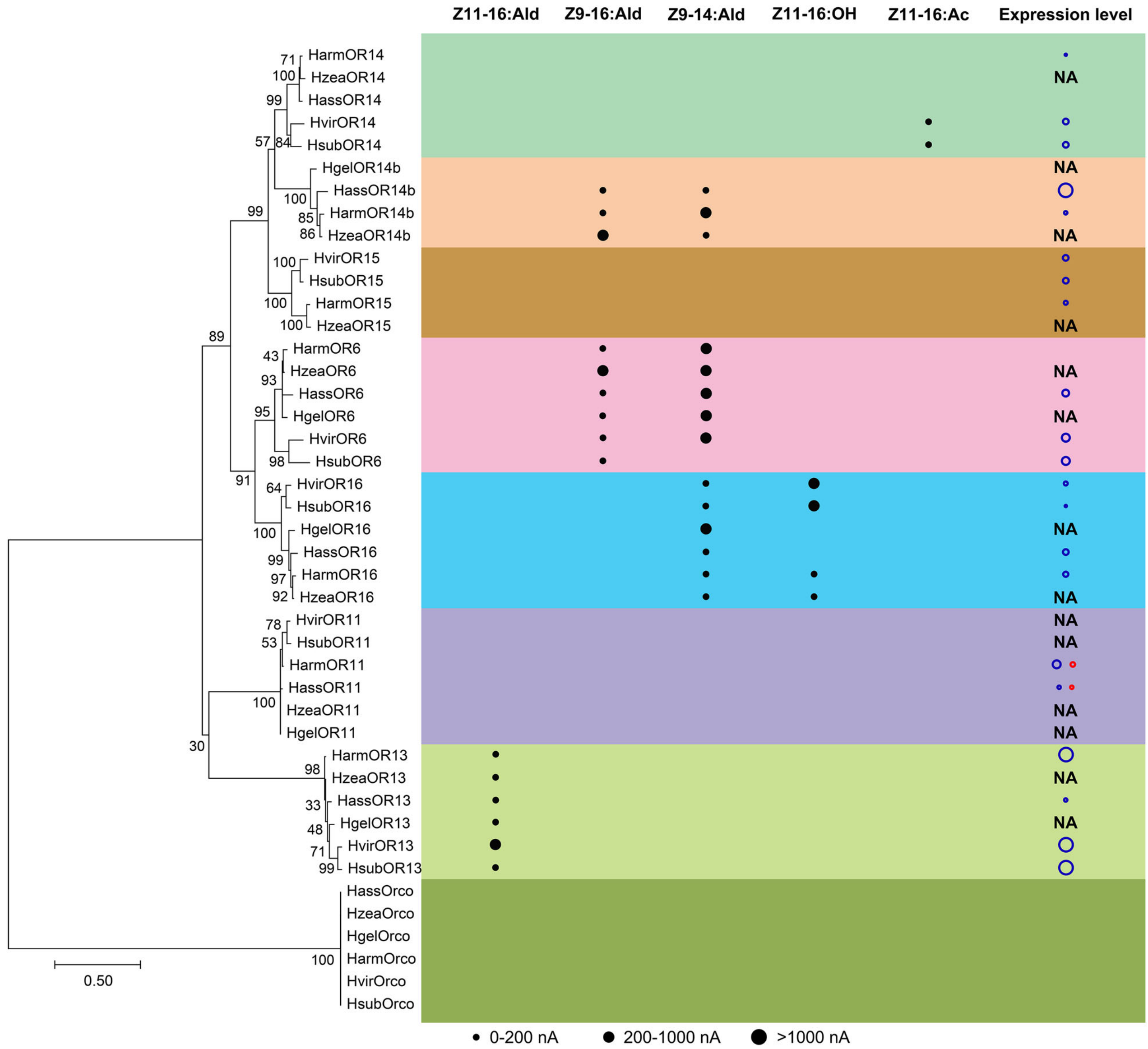

Fig. 3. Expression levels and functional comparison of pheromone receptors in six heliothine moth species. The size of each dot represents the magnitude of the response of each PR to its agonists (Table S1; from Wang et al., 2011; Liu et al., 2013b; Yang & Wang, 2020; Cao et al., 2021, 2023). Each PR clade is color-coded. Additionally, the relative expression level of each PR is normalized by designating the PR with the highest expression level as the largest circle in each species, and the relative expression of other PRs was calculated accordingly (Table S1; from Vásquez et al., 2011; Liu et al., 2013b; Yang et al., 2017; Yang & Wang, 2020). The blue and red cycles represent the expression levels of a gene in male and female adult antennae, respectively. NA, not available, indicates expression-level information of PRs has not been reported.

rapidly through this process, involving duplication, divergence, pseudogenization, and loss (Andersson et al., 2015; Benton, 2015). In heliothine moths, three PRs, *OR14*, *OR14b*, and *OR15*, have undergone birth-and-death evolution (Fig. 3). During the evolutionary history of these species, OR14 and OR15 have been retained in *Heliothis* species but exhibit varying degrees of loss in the *Helicoverpa* genus. Specifically, OR14 is lost in *H. gelotopoeon*, and OR15 is lost in both *H. gelotopoeon* and *H. assulta*. The loss of these receptors in *Helicoverpa* may reflect a lack of selective pressure. In contrast, OR14b appears to have been duplicated in the ancestor of *Helicoverpa*, likely due to strong selective pressure. Notably, unlike other heliothine species, *H. assulta* and *H. gelotopoeon* primarily use Z9-16:Ald as their major pheromone component, whereas Z9-16:Ald is a minor component in other species (Cork et al., 1992; Cork & Lobos, 2003). OR14b has been identified as the receptor for Z9-16:Ald in *H. assulta* (Yang et al., 2017), suggesting that the duplication of OR14b may have contributed to communication divergence in these moths. In the four *Spodoptera*

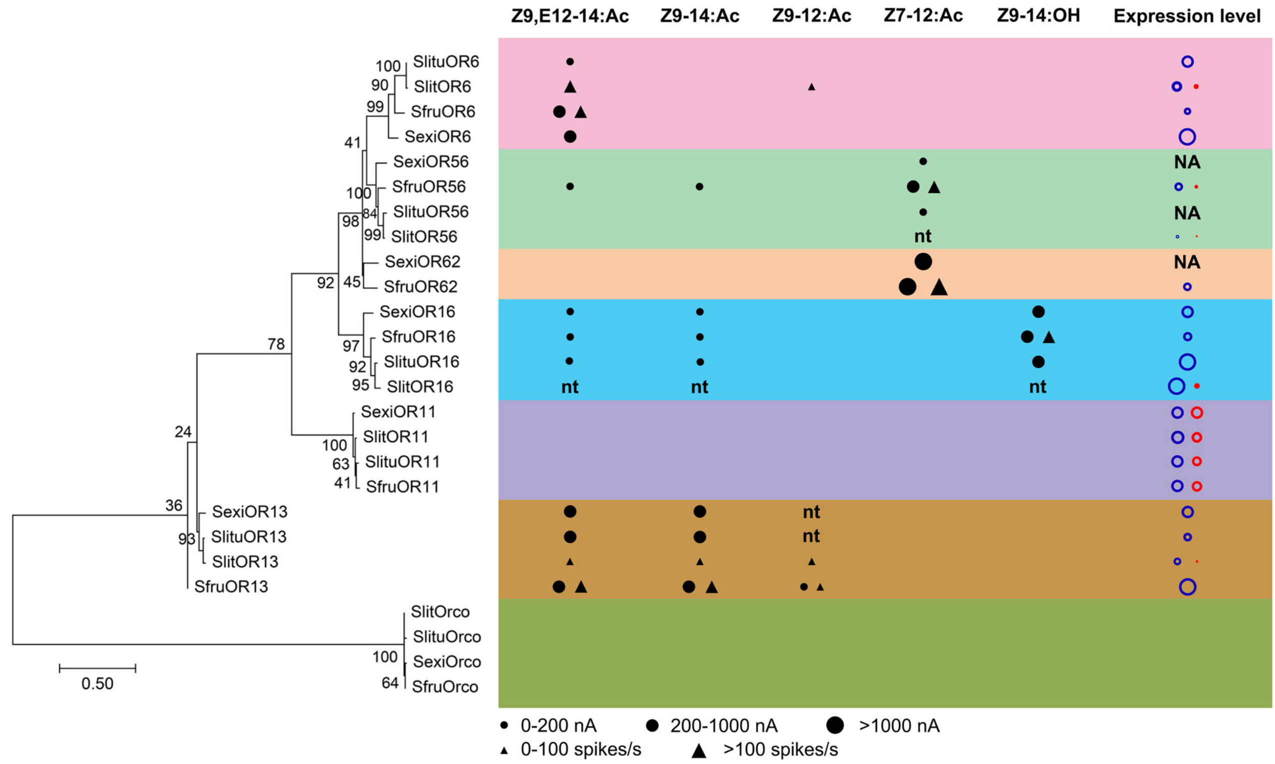

Fig. 4. Expression levels and functional comparison of pheromone receptors in four *Spodoptera* moth species. The size of each dot represents the magnitude of the response of each PR to its agonists (Table S1; from Liu et al., 2013a; de Fouchier et al., 2015; Zhang et al., 2015b; Guo et al., 2020; Guo et al., 2022; Zhang et al., 2023). Each PR clade is color-coded. Additionally, the relative expression level of each PR is normalized by designating the PR with the highest expression level as the largest circle in each species, and the relative expression of other PRs was calculated accordingly (Table S1; from Liu et al., 2013a; Zhang et al., 2015b; Walker et al., 2019; Guo et al., 2020; Guo et al., 2022; Zhang et al., 2023). The blue and red cycles represent the expression levels of a gene in male and female adult antennae, respectively. nt, not tested, indicates PRs has not yet been functionally tested; NA, not available, indicates expression-level information of PRs has not been reported.

species, only OR62 exhibits a birth-and-death evolutionary pattern. Given the evolutionary relationships among these species, OR62 appears to have been lost due to redundancy with OR56, as it is absent in *S. littoralis* and *S. litura*.

Second, variations in the expression levels of PR homologs significantly influence behavioral responses to pheromone blends. Generally, the receptor for the major pheromone component shows the highest expression level (Liu et al., 2013b; Yang et al., 2017). For example, OR13, which detects the major pheromone component Z11-16:Ald in *H. virescens*, *H. subflexa*, and *H. armigera*, exhibits the highest expression level among the PRs in these species (Fig. 3). Similarly, in *H. assulta*, OR14b, which recognizes Z9-16:Ald, also shows the highest expression level among all PRs. Although *H. armigera* and *H. assulta* share two pheromone components, Z11-16:Ald and Z9-16:Ald, their relative ratios differ, with 100:2.1 in *H. armigera* and 5.8:100 in *H. assulta* (Wang et al., 2005). In *H. armigera*, 53% of functional sensilla trichodea on the male antennae detect Z11-16:Ald, while only 10.3% respond to Z9-16:Ald. In contrast, in *H. assulta*, 30.9% of functional sensilla trichodea detect Z9-16:Ald, and 26.8% respond to Z11-16:Ald (Chang et al., 2016). Although both species can detect the two pheromone components, their behavioral responses differ. In wind tunnel assays, *H. assulta* males are strongly attracted to their own sex pheromone blend but show no response to that of *H. armigera* (Zhao et al., 2006). In the genus *Spodoptera*, significant divergence in pheromone compositions correlates with substantial variation in the expression levels of PR homologs (Fig. 4). In *S. exigua*, OR6 is the receptor for the major pheromone component and shows the highest expression level among PRs. In *S. frugiperda*, three PRs, OR13, OR16, and OR56, respond to the major sex pheromone Z9-14:Ac (Fig. 4). Among these, OR13, which exhibits the highest expression level (Guo et al., 2020, 2022), is likely the key receptor for Z9-14:Ac. In *S. litura* and *S. littoralis*, the PRs responsible for detecting their major sex pheromone, Z9,E11-14:Ac, belong to a novel PR clade rather than the traditional PR clade and are also highly expressed in male antennae (Bastin-Héline et al., 2019; Li et al., 2023). A similar pattern is observed in two species of the *Mythimna* genus. In *M. loreyi*, PR6 is specifically expressed in male antennae and exhibits the highest expression level among all PR genes. It also serves as the receptor for its major sex pheromone component, Z9-14:Ac (Wang et al., 2023, 2024a). In contrast, in *M. separata*, PR3 is the most highly expressed PR in male antennae and functions as the receptor for its major sex pheromone component, Z11-16:Ald (Jiang et al., 2019). In closely related species, differences in the expression levels of homologous genes involved in both sex pheromone biosynthesis and recognition are commonly observed across the genera discussed. However, the regulatory mechanisms underlying these differential expression patterns remain poorly understood. With advancements in genomics and gene editing technologies, unprecedented opportunities now exist to explore these mechanisms.

Third, amino acid mutations can alter the structure and function of PRs. Given the close relationship between receptor structure and function, mutations in key amino acids can lead to functional changes. These changes may involve alterations in tuning breadth, shifts in specificity, and modifications in sensitivity to ligands. However, in many cases, these changes are closely linked. OR16 shows functional changes in tuning breadth across the four *Helicoverpa* species (Fig. 3). In *H. armigera* and *H. zea*, OR16 responds to both Z9-14:Ald and Z11-16:OH, whereas in *H. assulta* and *H. gelotopoeon*, it is specifically tuned to Z9-14:Ald (Cao et al., 2023). This functional divergence of OR16 leads to distinct roles of Z11-16:OH in male behavior among the species (Quero & Baker, 1999; Chang et al., 2017). A single hydrophobic residue at position 66 contributes to the functional differentiation, likely through allosteric interactions across the species (Cao et al., 2023). Similarly, OR14b shows functional differentiation in *H. armigera* and *H. assulta* by shifting its specificity to Z9-14:Ald and Z9-16:Ald (Fig. 3). Two single-point mutations are responsible for this differentiation (Yang et al., 2017; Cao et al., 2023). In *H. virescens* and *H. subflexa*, OR6, which detects their respective minor pheromone components (Fig. 3), differs functionally due to a key amino acid mutation (Cao et al., 2021). In the genus *Mythimna*, *M. loreyi* and *M. separata* employ distinct pheromone signals, and three of the four functional PRs, PR1, PR3, and PR5, show functional differentiation (Jiang et al., 2019; Wang et al., 2021, 2024a). However, the mechanisms driving these functional shifts in these PRs remain unclear. In contrast, functional differentiation in PRs is less pronounced in the genus *Spodoptera*, despite significant differences in both sex pheromone components and the expression levels of PR homologs among the four species (Figs. 2, 4). Unlike *Heliothis* and *Helicoverpa*, where PRs within the traditional PR clade are sufficient to recognize female sex pheromones, multiple receptors in *Spodoptera* species, including those for major pheromone components, are clustered within the novel PR clade (Bastin-Héline et al., 2019; Li et al., 2023; Zhang et al., 2024). Among these species, PR orthologs exhibit varying degrees of sequence identity, with some being highly conserved while others showing functional divergence (Figs. 3, 4). Although the overall evolutionary pattern of these PRs is consistent with the species phylogeny, certain PR orthologs, such as OR11 and OR13, show topologies that do not align with it (Fig. 5). This incongruence suggests that some PR lineages have experienced distinct evolutionary constraints, highlighting their potential importance in pheromone recognition divergence. At present, however, such comparisons are most informative within closely related noctuid lineages, where homologous receptor clades and functional data can be compared directly. By contrast, broader cross-generic comparisons should be interpreted more cautiously because differences in pheromone composition, receptor repertoires, and the extent of functional characterization can obscure deeper ancestral-to-derived relationships. Thus, current data support lineage-specific evolutionary trajectories of male recognition rather than a single generalized pathway across Noctuidae.

Fourth, differences in the PR signaling pathway can lead to markedly different behavioral responses to the same chemical compound in closely related species. Once activated by chemical signals, insect ORs transmit sensory information to higher brain centers via projection neurons. The widespread projections of these neurons are organized according to the behavioral significance of the signals, including a spatial separation between signals that induce attraction versus inhibition. These projections also exhibit a unique capacity to switch the behavioral outcome based on

Fig. 5. Comparison of phylogenetic trees of ten noctuid moth species with their pheromone receptors. The methods used to construct the phylogenetic tree of PRs are similar as described in the legend for Fig. 2, related gene sequences are listed in Table S1 and Dataset S1.

the relative abundance of minor components in the chemical signal (Min et al., 2013; Kymre et al., 2021). Therefore, identifying the ligands for PRs alone cannot determine whether their effect on male behavior is positive or negative; further behavioral experimental data are needed to make this determination. For example, the OR14 homologs of *H. virescens* and *H. subflexa* both respond to Z11-16:Ac, a minor pheromone component of *H. subflexa*. This compound induces attractive behavior in *H. subflexa* males but inhibits *H. virescens* males (Teal et al., 1981; Vickers & Baker, 1997). The contrasting behavioral response may be due to differences in the projection region of OR14 neurons in the higher brain centers of these species. A similar pattern is observed in *Helicoverpa* species: OR13 retains the ability to recognize Z11-16:Ald in *H. gelotopoeon*, which is the only species in the genus not using Z11-16:Ald as a pheromone component. When added at a 1% level to its pheromone blend, male captures were significantly reduced, suggesting that the opposite behavioral response may result from a change in the projection site of OR13 neurons in the brain (Cork & Lobos, 2003).

In the genus *Spodoptera*, changes in the projection regions of homologous PRs leading to behavioral differences appear less pronounced than in *Helicoverpa* and *Heliothis*. This is because some receptors exhibit broader tuning spectra and can be activated by both behavioral agonists and antagonists, complicating the determination of their in vivo function. For example, OR13 in *S. frugiperda* is highly expressed in male antennae and can be activated by Z9,E12-14:Ac, Z9-14:Ac, and Z9-12:Ac (Guo et al., 2020, 2022). This is consistent with the number and function of Type I sensilla trichodea on male antennae (Wang et al., 2022). However, Z9-14:Ac, the major pheromone component, has opposite effects on male behavior compared to Z9,E12-14:Ac, a non-pheromone component (Mitchell et al., 1974; Wang et al., 2022). This situation appears paradoxical because, theoretically, a single PR should not be activated by both an agonist and an antagonist unless different concentrations of these compounds elicit distinct behavioral responses. Alternatively, these differences may reflect geographic variations in the behavioral responses of male moths to the same compound.

In addition to intrinsic changes in receptor repertoire, expression pattern, and ligand tuning, male pheromone preference in noctuid moths may also be modulated by ecological context. In *S. littoralis*, host-plant volatiles affect male responses to sex pheromone and reduce attraction to incomplete or heterospecific pheromone blends, thereby sharpening discrimination of conspecific signals (Borrero-Echeverry et al., 2018). These results indicate that plant odor and sex pheromone are not processed as fully independent cues during mate recognition. Instead, odor backgrounds can influence male recognition systems, with potential consequences for premating isolation and the stability of species-specific communication channels.

Taken together, pheromone recognition divergence in noctuid moths may arise through several possible mechanisms, including receptor gain and loss, expression divergence, changes in ligand selectivity, and ecological modulation of signal perception. At the same time, the current comparative framework remains incomplete. Many PRs in noctuid moths remain functionally uncharacterized and/or lack available expression level data. This incomplete coverage limits our ability to compare closely related species in a fully symmetrical manner. Moreover, most current functional assignments still depend on heterologous expression systems, especially *Xenopus* oocytes and transgenic *Drosophila* neurons. While these systems have greatly advanced receptor identification and ligand screening, they may not fully reflect native functions. Integrative in vivo approaches, particularly those combining CRISPR/Cas9-mediated functional studies with electrophysiological and behavioral analyses, will therefore be important for refining receptor assignments and improving evolutionary comparisons of pheromone recognition systems among closely related noctuid species.

4 Evolutionary hypotheses for signal-response divergence in noctuid moths

The comparative patterns summarized above indicate that pheromone divergence in noctuid moths is shaped by both changes in female blend composition and correlated changes in male recognition systems. This coupling provides a useful basis for evaluating hypotheses that have been proposed to explain how novel signal-response combinations are successfully established. Although these hypotheses were originally developed from broader studies across Lepidoptera, recent work on noctuid moths now allows a more explicit assessment of their relevance within this family.

The asymmetric tracking (AT) hypothesis suggests that selection on male preference is often stronger than selection on female signal production, such that males tend to track changes in female pheromone blends rather than constrain them (Phelan, 1992). Several noctuid examples support this view. *H. armigera* and *H. assulta* share the same two major pheromone components, Z11-16:Ald and Z9-16:Ald, but in opposite ratios (Wang et al., 2005). Comparative analyses of pheromone gland transcriptomes suggest that differential expression of the Δ11 and Δ9 desaturase genes is a major factor underlying this divergence in female signal output (Li et al., 2015, 2017). In males, the receptor tuned to the major pheromone component is also the most highly expressed PR in each species, with OR13 predominating in *H. armigera* and OR14b in *H. assulta* (Yang et al., 2017). This expression bias is further reflected in the relative abundance of functional sensilla responsive to the two compounds and in species-specific behavioral preference in wind-tunnel assays (Zhao et al., 2006; Chang et al., 2016). These results suggest that shifts in female pheromone composition can be accompanied by coordinated changes in the male peripheral recognition system.

Likewise, in *Spodoptera*, divergence in pheromone blends is accompanied not only by marked differences in the expression profiles of homologous PRs but also by an evolutionary case involving receptor duplication and tuning specialization. In most *Spodoptera* species, the major pheromone component is Z9-14:Ac, whereas in the sibling species *S. littoralis* and *S. litura*, the major component is Z9,E11-14:Ac, indicating a major shift in the communication system in their common ancestor. Functional analyses showed that OR5 orthologs in *S. exigua* and *S. frugiperda* are broadly tuned, whereas OR5 in *S. littoralis* and *S. litura* is narrowly tuned to Z9,E11-14:Ac and is strongly male-biased in expression. Ancestral sequence reconstruction further indicated that the receptor ancestral to the OR5/OR75 clade had a broader response spectrum that included this compound, whereas after gene duplication, one copy evolved narrow specificity and the other retained broader tuning. This pattern fits well with the asymmetric tracking hypothesis because broad ancestral male sensitivity could have permitted responses to a novel female-emitted signal before the later evolution of receptor specialization. The authors further identified multiple amino acid substitutions associated with the narrowing of the response spectrum, providing a detailed mechanistic example of how male recognition may evolve to stabilize a new pheromone channel in noctuid moths (Li et al., 2023).

The rare male hypothesis proposes that a novel female blend can persist if a small subset of males is able to respond to it, thereby allowing the novel female phenotype to escape immediate elimination. This idea remains plausible for noctuid moths, particularly in cases where intraspecific variation in male response and female blend has been reported, but direct genetic evidence is still limited. Likewise, the male mating mistakes hypothesis emphasizes imperfect male discrimination and the possibility that errors during orientation behavior may relax stabilizing selection on female sex pheromones. A relevant example comes from *Trichoplusia ni*, in which males exposed to a background cloud of behavioral antagonists showed reduced sensitivity to those compounds during orientation (Liu & Haynes, 1994). This result suggests that male preference can be context-dependent and somewhat plastic, potentially allowing transient behavioral windows through which divergent pheromone phenotypes may persist.

The directional selection hypothesis proposes that when two closely related species with similar communication channels occur sympatrically, communication interference can lead to reproductive character displacement, countering stabilizing selection (Cardé et al., 1977; Pfennig & Pfennig, 2009). For example, *H. subflexa* females produce a higher proportion (>5%) of acetate esters when in sympatry with *H. virescens*, reducing communication interference and driving directional selection (Groot et al., 2006).

The examples mentioned above indicate that signal-response divergence in noctuid moths is unlikely to be explained by any single mechanism. In some lineages, divergence appears to begin with shifts in female blend composition caused by changes in desaturase expression or mutation. In others, changes in receptor expression bias or tuning specificity may have an equally important role in shaping male preference. Ecological factors may further modulate these processes, especially in sympatric species where communication interference, geographic variation, and local selection can influence both pheromone composition and behavioral response.

Overall, for noctuid moths, these hypotheses are still working frameworks rather than fully validated explanations because direct experimental links between female signal change, male response evolution, and long-term lineage variation remain scarce. Fortunately, recent advances in noctuid genomics, functional analyses of biosynthetic enzymes, and receptor characterization have moved the field beyond purely theoretical conjecture. Noctuid moths now offer one of the best opportunities to connect the molecular basis of pheromone variation with the evolutionary processes that generate premating isolation and, ultimately, species divergence.

5 Conclusions and future directions

Taken together, the evidence reviewed here indicates that noctuid moths provide one of the best systems for understanding how species-specific communication systems diverge. Across representative noctuid lineages, including *Heliothis*, *Helicoverpa*, *Spodoptera*, and *Mythimna*, species-specific blends and male behavioral responses are often tightly matched, and even subtle differences in component identity or ratio can strongly affect mate attraction and premating isolation. Recent studies have further shown that these phenotypic differences are associated with genetic changes in biosynthetic or metabolic enzymes (Lassance et al., 2010; Li et al., 2017; de Fouchier et al., 2023) and pheromone receptors (Leary et al., 2012; Yang et al., 2017; Cao et al., 2023), moving the field beyond descriptive comparisons toward a more mechanistic understanding of signal-response evolution. Meanwhile, the available evidence remains uneven across Noctuidae, with some genera being studied in considerable detail whereas many noctuid lineages remain only weakly characterized.

Several directions seem particularly important for future work on noctuid moths.

(1) Functional validation of candidate sex pheromone components in additional noctuid lineages. Although pheromone compounds have been identified for many noctuid species, behavioral validation remains incomplete in a substantial number of taxa. More electrophysiological and behavioral studies are needed to distinguish true sex pheromone components from gland-associated compounds and to determine how individual components contribute to attraction, antagonism, and premating isolation in closely related noctuid species.
(2) Dissection of the genetic architecture of female pheromone divergence in noctuid moths. Recent work in *Helicoverpa* and *Heliothis* has shown that changes in desaturases, lipases, esterases, and other enzymes can reshape pheromone blends, but the broader regulatory architecture remains poorly understood. Future studies should identify the transcription factors, regulatory elements, and pathway interactions that control species-specific pheromone production, especially in closely related noctuid taxa with contrasting blend ratios or pheromone types.
(3) In vivo analysis of male recognition systems in representative noctuid species. Progress in receptor deorphanization has greatly advanced our understanding of peripheral pheromone recognition, but many receptors in noctuid moths remain functionally uncharacterized, and heterologous assays may not fully reflect their

roles in vivo. Combining receptor functional assays with CRISPR/Cas9-mediated gene editing, sensillum-level physiology, and behavioral analysis will be essential for determining how changes in receptor expression, receptor sequence, and neural representation alter male preference in noctuid moths.

(4) Integration of ecological context into studies of communication divergence. Geographic isolation, sympatry, host-associated differences, and environmental stress may all influence pheromone composition and male response in noctuid moths. Future field-based studies should therefore examine how ecological conditions affect the stability of species-specific communication systems and whether environmental heterogeneity promotes local differentiation or the maintenance of alternative pheromone phenotypes within noctuid populations.

(5) Comparative evolutionary analyses across Noctuidae. One major challenge is to determine at what phylogenetic scale current data allow reliable inference of ancestral versus derived communication traits. At present, the strongest evolutionary inferences come from closely related noctuid lineages in which pheromone composition, receptor evolution, and species relationships can be compared directly, whereas deeper family-level reconstructions remain limited by sparse functional and biochemical data. Addressing this question will require denser phylogenetic sampling across noctuid moths, especially beyond the currently well-studied heliothines and *Spodoptera*, together with comparative analyses that link pheromone composition, biosynthetic pathways, receptor evolution, and species relationships. Such work will help clarify whether similar communication phenotypes in different noctuid clades arise through shared ancestry, convergence, or repeated recruitment of similar molecular mechanisms.

In summary, noctuid moths offer an exceptional opportunity to connect molecular changes in pheromone communication with broader evolutionary processes. A more complete synthesis of genomic, molecular, ecological, neurobiological, and behavioral data within Noctuidae will clarify when communication divergence is sufficient to maintain reproductive isolation and when it may scale up to broader evolutionary divergence. At present, the clearest evolutionary trajectories are those reconstructed within closely related noctuid lineages, whereas broader family-level patterns remain an important goal for future comparative work. This research will also continue to inform innovative pest-management strategies that target moth mating behavior through pheromone manipulation.

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Author Contributions

All authors contributed to defining the scope of this paper, conducting literature-based analysis on the topic, writing, and editing the manuscript.

Conflicts of Interest

The authors declare no conflict of interest.

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Supplementary Material

The following supplementary material is available online for this article at http://onlinelibrary.wiley.com/doi/10.1111/jse.70085/suppinfo:

Dataset S1. One-to-one orthologous genes for phylogenetic analysis of ten noctuid moth species.

Table S1. Data sources used to generate Figs. 2–5.